LABORATORY PROJECT REPORT

Personal Portfolio, Resume & Bio-Data Web Application

Academic Implementation, Architecture Design & Cloud Deployment Documentation

COURSE / LAB	Web Technology / Internet Technology Laboratory (Assignment 1)	CANDIDATE NAME	Cynthia Alexander
DEGREE & COLLEGE	B.E. (Software Engineering), Jain College, Hubli, Karnataka	LIVE DEPLOYMENT	assignment-1-topaz-alpha.vercel.app
VERSION CONTROL	github.com/cynthiaalexandergeorge-create/assignment-1	SUBMISSION DATE	November 2026

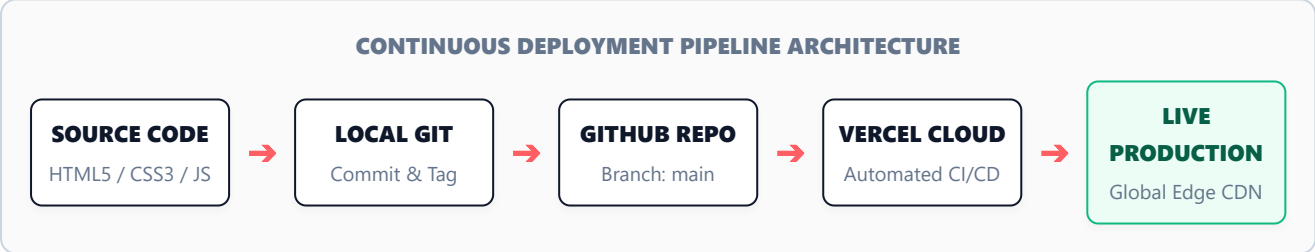
1. Executive Summary

This project delivers a multi-page personal portfolio, curriculum vitae, and bio-data web platform designed to satisfy academic web development laboratory specifications. The web platform establishes a professional online digital identity for **Cynthia Alexander**, an engineering student at Jain College, Hubli.

Starting from foundational laboratory reference specifications, the system was re-architected following contemporary UI/UX design standards, semantic HTML5 structure, modern CSS3 layout systems (CSS Grid and Flexbox), vanilla JavaScript for DOM manipulation, and pure SVG vector graphics. The completed codebase is version-controlled with Git, maintained on GitHub, and hosted on the Vercel Edge Network.

2. Technology Stack & Deployment Architecture

LAYER / COMPONENT	TECHNOLOGY	FUNCTIONAL ROLE
Document Structure	HTML5 (Semantic)	Provides structural semantic markup, accessibility (A11y), and form controls.
Presentation Layer	CSS3 (Vanilla)	Custom CSS design tokens, CSS Grid, Flexbox, keyframe animations, and media queries.
Behavioral Logic	JavaScript (ES6+)	Mobile navigation toggle, interactive form validation, and event handling.
Vector Iconography	Inline SVG (Scalable Vector Graphics)	Resolution-independent icons and animated rotating vector badge.
Photography Assets	Web-Optimized JPEG	High-resolution studio portfolio and portrait photography.
Version Control	Git & GitHub	Branch management (`main`), distributed versioning, and remote sync.
Hosting & CI/CD	Vercel Edge Network	Automated webhook continuous deployment, global CDN, and automated SSL.



3. Web Platform Modules & Specifications

3.1 Home Page (index.html)

- **Header & Navigation:** Sticky desktop and mobile responsive navigation bar with brand monogram badge (`C`), page routing links, and high-contrast call-to-action button.
- **Hero Showcase:** Prominent presentation featuring an electric-lime tag (`Hello I Am`), bold headline (`I'm Cynthia`), professional summary, and call-to-action routing.
- **Circular Vector Rotating Badge:** Pure SVG `<textPath>` curving smoothly along a circular radius (`SOFTWARE ENGINEERING • CYNTHIA •`) executing continuous 360° keyframe animation around studio portrait photography.
- **Dark Feature Ticker:** Obsidian-black banner with electric-lime sparkle vectors highlighting competencies in Software Engineering, AI & Federated Learning, Web Architecture, and Problem Solving.
- **Embedded About Preview:** Introductory biography preview guiding visitors to the complete profile.

3.2 About Me (about.html)

- **Narrative Statement:** In-depth academic narrative highlighting software engineering coursework at Jain College, focus on clean architecture, and passion for distributed artificial intelligence systems.
- **Visual Anchor:** Clean monochrome portrait photography.
- **Core Identity Badges:** Structured capsule tags denoting candidate attributes: `[Software Engineering Student]`, `[Tech Enthusiast]`, `[Problem Solver]`, `[Quick Learner]`, and `[Hubli, Karnataka]`.

3.3 Curriculum Vitae / Resume (resume.html)

- **Personal Information:** Direct communication points (Email: `cyntalex@gmail.com`, Phone: `+91 8972639101`, Location: `Hubli, Karnataka`).
- **Career Objective:** Professional aspiration statement in enterprise software engineering.
- **Educational Qualifications:**

LEVEL	INSTITUTION	COMPLETION YEAR	PERCENTAGE
B.E. (Software Engineering)	Jain College	2026	**85%**
PUC (Pre-University)	Adarsh College	2022	**80%**
SSLC (10th Standard)	Government School	2020	**90%**

- **Technical Competencies:** C Programming, HTML & CSS, JavaScript, Python, Database Management, and AI & Federated Learning.
- **Project Portfolio:**
  - *Student Management System:* Comprehensive academic record and grading platform.
  - *Food Waste Management System:* Platform connecting surplus food supply with community networks to minimize waste.
  - *AI in Federated Learning:* Decentralized privacy-preserving machine learning framework.

- **Achievements & Hobbies:** Technical quiz competitions, web development workshops, coding competitions, literature reading, and music.

3.4 Official Bio-Data (biodata.html)

- **Institutional Record Format:** Detailed key-value table recording Full Name, Date of Birth (`01 January 2006`), Gender (`Female`), Nationality (`Indian`), Languages (`English, Kannada, Tamil, Hindi`), Residential Address, and Contact numbers.
- **Legal Declaration:** Formal statement certifying truthfulness and factual accuracy.
- **Authentication Block:** Place (*Hubli*), Date (*25/09/2026*), and formal signature block.

3.5 Contact Module (contact.html)

- **Direct Channel Cards:** Contact channels for Email, Telephone, Geographic Address, and Internship Availability Status.
- **Interactive Message Form:** Client-side validated form fields (Name, Email, Subject, Message) coupled with instant feedback confirmation banner on submission.

4. Key Design & Engineering Techniques

> **Pure SVG Vector Engineering:** Casual emoji characters were systematically eliminated in favor of clean, resolution-independent SVG vector iconography. Stroke-based geometry (`viewBox="0 0 24 24"`, `stroke-width="2"`) ensures sharp rendering across high-DPI retina displays and seamless theme color inheritance via `currentColor`.

- **CSS Design Tokens:** Unified stylesheet utilizing variables for brand colors (`--primary: #ff5e62`, `--accent-lime: #ccff00`, `--dark-bg: #090a0f`) and typography scales.
- **Responsive Viewport Breakpoints:** Dynamic media queries at `900px` and `650px` ensuring accessibility across smartphones, tablets, and desktop workstations.
- **Semantic Markup:** Proper use of `<header>`, `<nav>`, `<main>`, `<section>`, and `<footer>` for optimal screen reader accessibility and SEO.

5. Live Verification & Endpoint Test Results

PAGE ROUTE	PRODUCTION ENDPOINT	HTTP STATUS	AUDIT RESULT
Home	`https://assignment-1-topaz-alpha.vercel.app/`	**200 OK**	Rendered successfully with all styles and assets.
About Me	`https://assignment-1-topaz-alpha.vercel.app/about.html`	**200 OK**	Full profile narrative and portrait operational.
Resume / CV	`https://assignment-1-topaz-alpha.vercel.app/resume.html`	**200 OK**	Complete education, skills, and projects verified.
Bio-Data	`https://assignment-1-topaz-alpha.vercel.app/biodata.html`	**200 OK**	Personal details table and declaration confirmed.
Contact	`https://assignment-1-topaz-alpha.vercel.app/contact.html`	**200 OK**	Contact channels and interactive form verified.

Verified and submitted by:

Cynthia Alexander

Department of Software Engineering, Jain College

Location: **Hubli, Karnataka**

Date: **07 November 2026**